1

Day: Control system Date:

Assignment:

Q:1

$$G(s) = \frac{K(s+6)}{(s+2)(s+3)(s+5)}$$

along $\zeta = 0.707$ which is 135°

so $s = -2.32 + j2.32$, $K = 4.6045$

$$T_p = \frac{\pi}{2.32} = 1.354.$$

$$T_s = \frac{4}{2.32} = 1.724 \text{ second.}$$

$$K_p = \frac{4.6045}{30} = 0.153$$

$$\%OS = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} * 100 = 4.3\%$$

$$\omega_n = \sqrt{(2.32)^2 + (2.32)^2} = 3.28$$

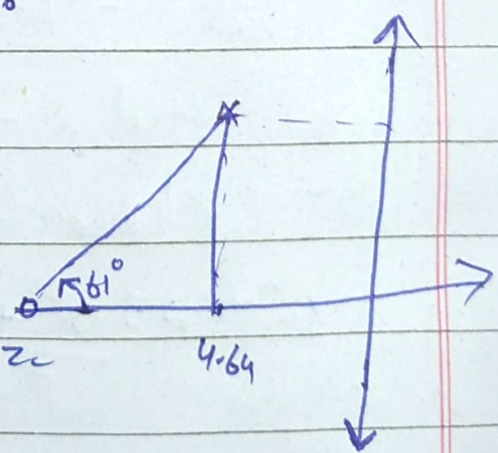
Compansated & reduce by 2

then close loop $-4.64 \pm j4.64$

angle sum 119°

$= 180^\circ - 119 = 61^\circ$

$z_c = \tan(61^\circ) = 7.21$



Then By adding compansator zero.

Gain can be 4.77

$$K_p = \frac{4.77 * 6 * 7.21}{2 * 3 * 5} = 6.88$$

Day: 2 Date:

$T_s = \frac{4}{4.64} = 0.86$ sec

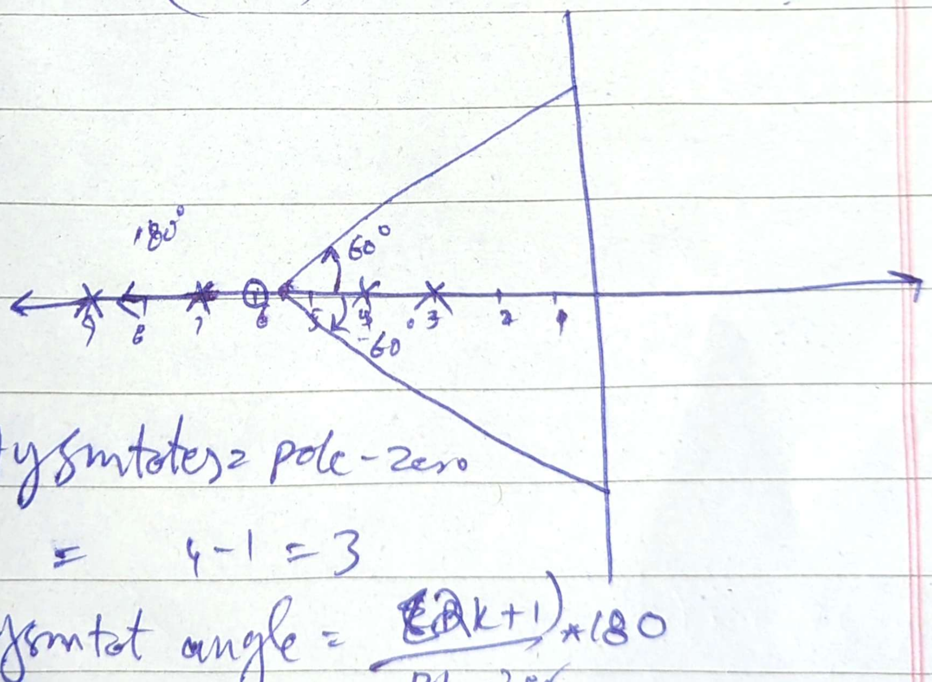
$T_p = \frac{\pi}{4.64} = 0.67$ second.

$\%OS = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100 = 4.33\%$

$\omega_n = 6.56$ rad/second

There is a steady state error and no effective pole & zero cancellation.

Q:02 $\dfrac{K(s+6)}{(s+3)(s+4)(s+7)(s+9)}$



Asymptotes = pole - zero

$= 4 - 1 = 3$

Asymptot angle $= \dfrac{(2k+1)}{\text{pole}-\text{zero}} \times 180$

$= 60°, 180°, 300°/-60°$

Day: Date:

$$\text{Centeroid} = \frac{(3+4+7+9)-6}{3}$$

$$= \frac{17}{3} = 5.6$$

For Break in & Break away.

$$1+\frac{k(s+6)}{(s+3)(s+4)(s+7)(s+9)} = 0$$

$$=(s+3)(s+4)(s+7)(s+9)+k(s+6)=0$$

$$k = -\left[\frac{(s+3)(s+4)(s+7)(s+9)\cdot 1}{(s+6)}\right]$$

By differentiating.

$$\frac{dk}{ds} = \frac{d}{ds}\left[\frac{(s+3)(s+4)(s+7)(s+9)}{(s+6)}\right].$$

$$s = -7.973, \ -3.452$$

and $-5.934 \pm j1.099$

So characteristic equation.

$$s^4+23s^3+187s^2+633s+756+ks$$
$$+6k$$

$$s^4+23s^3+187s^2+(633+k)s+6k+756$$

Day: Date:

s^4	1	187	$756+6k$	0
s^3	23	$(633+k)$	0	
s^2	$\frac{4301-(633+k)}{23}$	$756+6k$		
s^1	$\frac{(3668-k)}{23}$			
s^0				

$$s^0 \quad (633+k)\frac{(3668-k)}{23} - (23)(756+6k)$$

Also

$$\zeta = \cos\theta \rightarrow \theta = \cos^{-1}(\zeta)$$

$$\boxed{\theta = 36.87^\circ}$$

$$\theta = \tan^{-1}\left(\frac{\omega d}{\sigma}\right)$$

$$\omega d = \sigma \tan^{-1}\theta$$

$$s = -\sigma \pm j\omega_d$$

$$s = -\sigma \pm j\omega(0.75)$$

assume $\sigma = 2.5$

$$\boxed{s = -2.5 + j1.875}$$ dominant pole

Day: Date:

$$K = \frac{|(s+3)(s+4)(s+7)(s+9)|}{|s+6|}$$

Put $s = -2.5 + j\,1.875$

$$|s+6| = \sqrt{(3.5)^2 + (1.875)^2}$$

$$= 3.97$$

By calculating distance others are same

$$K = \frac{(1.94)(2.40)(4.88)(6.77)}{3.97}$$

$$\boxed{k = 38.7}$$

Now For Compensator.

$T_s = 1$, $\zeta = 0.8$, $s = -4.5$

$$\omega_n = \frac{4}{\zeta T_s} = \frac{4}{0.8 \times 1} = \boxed{5 = \omega_n}$$

dominent pole $\rightarrow s = -4 \pm j3$

$\angle$ zero $-$ $\angle$ poles $= -180°$

Day: Date:

$\angle(s+4.5) + \angle(s+6) - 2\cdot 74.39 = -180^\circ$

$\angle(s+4.5) + \angle(s+6) - \angle(s+P_c) - \angle(s+3) -$
$\angle(s+4) - \angle(s+7) + \angle(s+7)$

By putting $s = -4 \pm 3j$

$136.85^\circ - \angle(s+P_c) - \overset{2.74.39^\circ}{\uparrow} = -180^\circ$

$-137.54^\circ - \angle(s+P_c) = -180^\circ$

$\angle(s+P_c) = 42.46^\circ$

$\angle(-4+3j+P_c) = 42.46^\circ$

So:

$$\tan(42.46) = \frac{3}{P_c - 4}$$

$$P_c - 4 = \frac{3}{0.916}$$

$$\boxed{P_c = 7.275}$$

seconder order:

→ when zero's do not effect response.

→ dominent pole closer to jw axis.